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ORIGINAL RESEARCH

Prevalence of Cardiovascular Disease and Risk Factors Among Former National Football League Players

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ABSTRACT

BACKGROUND Cardiovascular disease (CVD) is the leading cause of death worldwide, but prevalence estimates in former professional athletes are limited.

OBJECTIVES HUDDLE (Heart Health: Understanding and Diagnosing Disease by Leveraging Echocardiograms) aimed to raise awareness and estimate the prevalence of CVD and associated risk factors among members of the National Football League (NFL) Alumni Association and their families through education and screening events.

METHODS HUDDLE was a multicity, cross-sectional study of NFL alumni and family members aged 50 years and older. Subjects reported their health history and participated in CVD education and screening (blood pressure, electrocardiogram, and transthoracic echocardiogram [TTE] assessments). Phone follow-up by investigators occurred 30 days postscreening to review results and recommendations. This analysis focuses on former NFL athletes.

RESULTS Of 498 participants screened, 57.2% (N = 285) were former NFL players, the majority of whom were African American (67.6%). The prevalence of hypertension among NFL alumni was estimated to be 89.8%, though only 37.5% reported a history of hypertension. Of 285 evaluable participants, 61.8% had structural cardiac abnormalities by TTE. Multivariable analysis showed that hypertension was a significant predictor of clinically relevant structural abnormalities on TTE.

CONCLUSIONS HUDDLE identified a large discrepancy between participant self-awareness and actual prevalence of CVD and risk factors, highlighting a significant opportunity for population health interventions. Structural cardiac abnormalities were observed in most participants and were independently predicted by hypertension, affirming the role of TTE for CVD screening in this population aged older than 50 years. (Heart Health: Understanding and Diagnosing Disease by Leveraging Echocardiograms [HUDDLE]; NCT05009589) (J Am Coll Cardiol 2024;83:1827–1837) © 2024 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

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ABBREVIATIONS AND ACRONYMS

BP = blood pressure

CV = cardiovascular

CVD = cardiovascular disease

ECG = electrocardiogram

HF = heart failure

HTN = hypertension

LVMI = left ventricular mass index

NFL = National Football League

TTE = transthoracic echocardiogram

Cardiovascular disease (CVD) remains the leading cause of mortality worldwide, despite improvements in CVD management and care.[1-3] Reasons for this may include an aging and growing global population, an underestimation of CVD and risk factor prevalence with an associated lack of engagement on an individual level, and a failure to diagnose and intervene early. High systolic blood pressure (BP) is the number one risk factor contributing to premature CVD mortality worldwide, yet an increasing number of people are living with suboptimal BP control.[2,3]

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CVD screening may help reduce the burden of undiagnosed disease and provide opportunities to slow progression and reduce costs; however, current guidelines are largely based on risk scores, which often do not provide a complete clinical picture and may overlook subpopulations who could benefit from basic cardiovascular (CV) testing.[4,5] One such population is athletes. Athletes are thought to be paragons of health because of their perceived healthy living habits and regular exercise, but they may be at significant risk for CVD both at a younger age and when they retire, and studies have shown that the risk may be particularly elevated among American-style football players.[6] Additionally, most studies report on current professional athletes, whereas data for retired athletes are lacking.

HUDDLE (Heart Health: Understanding and Diagnosing Disease by Leveraging Echocardiograms; NCT05009589) aimed to assess the prevalence of CVD and associated risk factors among a representative population of former National Football League (NFL) players, coaches, and family members through noninvasive CVD screening. The study also sought to quantify the knowledge gap of actual vs perceived disease, engage with community members through education, and identify opportunities for inclusive screening to mitigate the consequences of undiagnosed and undertreated CVD.

METHODS

STUDY DESIGN AND PATIENT POPULATION. HUDDLE was a multicity, cross-sectional study that evaluated clinical and lifestyle characteristics among members of the NFL Alumni Association, including former NFL players and coaches as well as their extended family. The focus of this paper will be on former NFL athletes. The study was sponsored by Edwards Lifesciences and aimed to assess the prevalence of CVD and related risk factors in an understudied population through cardiac screening and participant questionnaires. The sponsor funded and was involved in all trial-related activities, including study design, data management, and statistical analyses. A central Institutional Review Board approved the protocol, and all aspects of the study were performed in accordance with the ethical principles established by the Declaration of Helsinki. Written informed consent was obtained for all participants before screening activities.

A total of 9 screening events (including a small pilot event to assess feasibility) were held across 8 U.S. cities between August 2021 and August 2022 (Supplemental Table 1). Study participants were first identified through their local NFL Alumni Association Chapters and recruited through email and phone outreach. All recruitment activities were conducted approximately 1 to 2 months before each screening event. Participants were eligible for enrollment if they were an NFL alumni or family member, were aged ≥50 years, and provided written informed consent by the time of screening. Participants were excluded if they had a current diagnosis of COVID-19 or condition that would interfere with obtaining an echocardiogram of the heart, including but not limited to chronic obstructive pulmonary disease, a left-side pectoral or breast implant, or a body mass index of ≥50 kg/m^2.

SCREENING PROCEDURES. All screening events were organized by the sponsor and overseen by at least 2 physicians. The screening events comprised 3 elements: health history collection, CV screening, and CVD education (Supplemental Figure 1). Before screening, enrolled participants were asked to self-report their personal health history either through a phone interview with a sponsor representative or by directly entering it into an online patient portal managed by the sponsor. The health history form included questions regarding basic demographic and lifestyle information, current medications, and known medical history.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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TABLE 1 Baseline Characteristics and Participant Demographics of Former NFL Players (N = 285)

Age, y	63.1 ± 7.27 (285)
Race	
American Indian or Alaska Native	1.1 (3/284)
Asian	0.0 (0/284)
African American or Black	67.6 (192/284)
Native Hawaiian or other Pacific Islander	1.4 (4/284)
White	26.4 (75/284)
Other	2.5 (7/284)
Multiple	1.1 (3/284)
Ethnicity	
Hispanic or Latino	3.6 (10/281)
Not Hispanic or Latino	86.5 (243/281)
Unknown or prefer not to say	10.0 (28/281)
Body mass index, kg/m^2	31.2 ± 4.66 (285)
Body surface area, m^2	2.4 ± 0.23 (285)

Values are mean ± SD (N) or % (n/N); denominators may vary because not all participants answered every question.
NFL = National Football League.

At each event, participants underwent the following noninvasive CV assessments: BP, 12-lead electrocardiogram (ECG), and transthoracic echocardiogram (TTE). The screening areas were operated by a central group of independent sonographers, all of whom were trained on the study screening criteria and equipment provided by the sponsor. To ensure uniformity in measurements, every event was equipped with the same portable ECG (Philips Pagewriter DXL TC30), echocardiography (GE Vivid iq Premium Ultra Edition v204), and automated professional BP (Omron Intellisense HEM-907XL) machines on which the sonographers were trained. ECG and TTE assessments were performed per current guidelines.[7,8] BP measurements were taken from a seated, resting position after the ECG was performed. If "white-coat" hypertension (HTN) was suspected, additional readings were taken at the sonographer or physician's discretion throughout the screening event. The average BP measurement was recorded for participants who received 2 or more readings.

The overseeing physicians consulted with participants during screening and reviewed each assessment for quality and any critical findings. An independent central core lab analyzed all deidentified ECG tracings and TTE images. The third component of the screening events was an information session and physician-led panel about CVD, its risk factors, common symptoms, and prevention.

FOLLOW-UP AND DATA MANAGEMENT. Follow-up phone visits were conducted by one of the principal physicians at 30 days postscreening. Investigators reviewed the questionnaire and CVD screening results with each participant and provided recommendations for further follow-up based on clinical judgment. Referrals to local providers were offered if the participant was not already under the care of a specialist. Results from each participant's medical history, screening tests, and follow-up calls were deidentified and entered in a secure electronic database managed by the sponsor.

ESTIMATION OF CVD AND RISK FACTOR PREVALENCE. Readings for BP were categorized according to the 2017 American College of Cardiology/American Heart Association clinical guidelines for the assessment of BP in adults, with HTN defined as a BP of ≥130/80 mm Hg at screening.[9] The prevalence of HTN was estimated as the proportion of subjects who either reported a history of HTN or were hypertensive at screening to all screened participants. Abnormal ECG findings were defined as the presence of ≥1 prognostic findings, which were selected based on clinical expertise (Supplemental Table 2). Structural heart abnormalities per TTE were identified using sex-specific thresholds according to American Society of Echocardiography guidelines and are detailed in Supplemental Table 3.[10]

STATISTICAL ANALYSIS. The analytic sample consisted of former NFL players who underwent screening assessments. No prespecified hypothesis testing or power-based sample size calculations were performed. Summary statistics are presented as count and percentage for categorical variables and mean ± SD for continuous variables. A multivariable logistic regression model was used to identify participants at high risk for undiagnosed structural heart disease detected by TTE. Clinically relevant variables with prognostic implications for mortality were selected a priori based on clinical expertise of the principal physicians and are provided in Supplemental Table 4. Covariates in the multivariable model were selected based on their significance in the univariable model at the 0.1 significance level. All analyses were performed using SAS version 9.4.

RESULTS

PARTICIPANT FLOW AND HEALTH HISTORY QUESTIONNAIRE. Of 552 participants enrolled in HUDDLE, 498 (90.2%) attended 1 of 9 screening events, of whom 285 were former NFL players (Supplemental Figure 2). Baseline participant demographics are detailed in **Table 1**. The majority of

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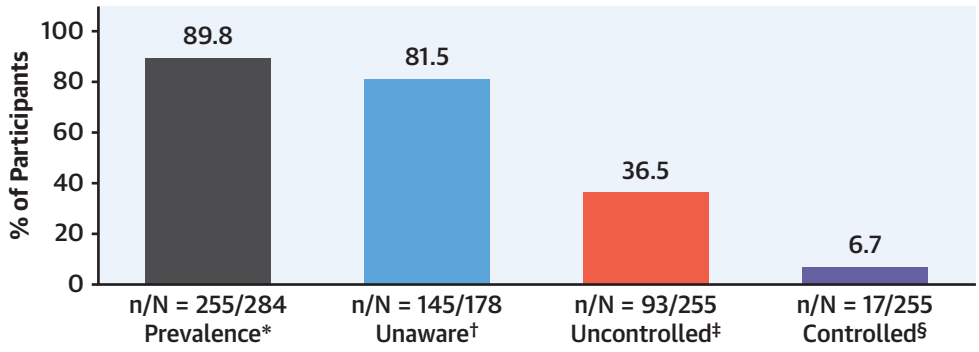
FIGURE 1 HTN Prevalence and Control



The prevalence of HTN among former professional athletes as well as levels of awareness and control. Despite reporting a history of HTN, 1 participant was excluded from this analysis because his BP was unmeasurable at screening. (His arm was too large for available cuff sizes.) *Defined as the proportion of participants who reported a history of HTN or had a BP of ≥130/80 mm Hg at screening or who reported taking a medication used to treat HTN over all participants screened. †Defined as the proportion of participants who had a BP ≥130/80 mm Hg at screening over participants who did not report a history of HTN. ‡Defined as the proportion of participants who reported a history of HTN and who had a BP of ≥130/80 mm Hg at screening over the prevalence (*). §Defined as the proportion of participants who had a BP of <130/80 mm Hg at screening over the prevalence (*). BP = blood pressure; HTN = hypertension.

participants were African American (67.6%), and the average age was 63.1 years, with a range of 50 to 82 years. The mean body mass index among former NFL players was 31.2 kg/m^2 and ranged from 21.0 to 46.7 kg/m^2. Information on lifestyle habits was collected, with most participants reporting moderate to rigorous weekly exercise (Supplemental Table 5).

Known CVD and risk factor history was collected as part of the participant questionnaires, as was non-CVD history (Supplemental Table 6). The 2 most prevalent known CVD risk factors were HTN and high cholesterol (37.5% and 32.6%, respectively). Only 6.3% reported having coronary artery disease, and 2.1% had a known history of heart failure (HF). The most frequently reported non-CVD condition was muscle pain or body aches (47.7%), followed by frequent headaches/dizziness (22.5%). There were 40 participants who reported no known medical history (no CVD or non-CVD). Participants were also asked to list current medications, which were later categorized by type; there were 169 participants (59.3%) who reported taking CV medications (Supplemental Table 7).

TABLE 2 Blood Pressure Screening Results of Former NFL Players (N = 284)[a]

Normal (<120/80 mm Hg)	4.6 (13)
Elevated (≥120/80 mm Hg and <130/80 mm Hg)	11.6 (33)
Hypertensive (≥130/80 mm Hg)	83.8 (238)
≥130/80 mm Hg and <140/90 mm Hg	78
≥140/90 mm Hg and <150/100 mm Hg	61
≥150/100 mm Hg and <200 mm Hg systolic	97
≥200 mm Hg systolic	2

Values are n or % (n). Resting blood pressure was measured from a seated position, and values were averaged if ≥2 readings were taken. [a]Blood pressure was unmeasurable in 1 participant because his arm was too large for available cuff sizes; however, he reported a history of hypertension and taking antihypertensive medication.

NFL = National Football League.

CVD AND RISK FACTOR PREVALENCE. The prevalence of HTN among the former NFL players was estimated to be 89.8%; only 6.7% of those participants recorded a normal BP reading at the screening event (**Figure 1**). Overall, 83.8% of participants had a BP reading of ≥130/80 mm Hg at screening (**Table 2**). One participant was unable to have his BP read because his arm was too large for available cuff sizes; however, he had a known history of HTN and was reportedly taking an angiotensin-converting enzyme inhibitor for BP control. Several participants at screening were found to be in hypertensive crisis and were advised to go to the emergency department or urgent care following the screening event, including participants who reported no known history of HTN. The HTN prevalence of 89.8% when contrasted with the 37.5% of subjects who reported a history of HTN highlights a huge knowledge gap among screened participants.

Abnormal ECG findings were identified in 46.1% of former NFL players based on the definitions in Supplemental Table 2 (**Central Illustration**). The presence of 1 or more structural heart abnormalities was

CENTRAL ILLUSTRATION **Key Insights From HUDDLE**

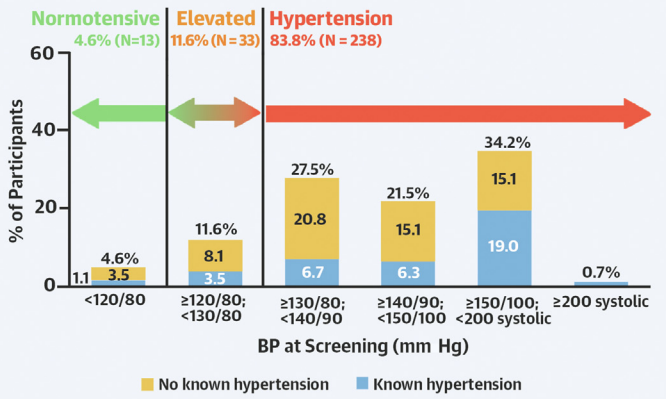
Study Design and Follow-Up

Multicity, cross-sectional study to assess the prevalence of CVD and related risk factors in former NFL players aged ≥50 years

Participants were screened using health history questionnaires and 3 noninvasive cardiac assessments: BP, ECG, and TTE

100% of former players screened (N = 285) completed 30-day phone follow-up with study investigators

Knowledge Gap: BP at Screening by History



CVD and Risk Factor Prevalence

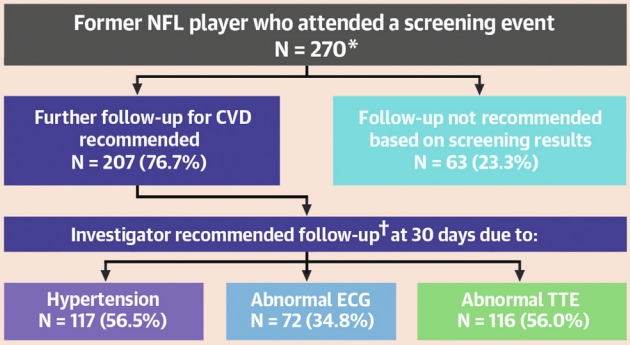
Hypertension (includes BP screening and health history) → 89.8% (255/284)

Abnormal ECG (12-lead) → 46.1% (131/284)

Structural Abnormalities on TTE → 61.8% (176/285)

Structural abnormalities were present in 64.0% (48/75) of White and 61.0% (128/210) of non-White participants

Investigator Recommendations



Hypertension was independently predictive of a structural abnormality on TTE (OR [95% CI]: 2.02 [1.03-3.96])
P = 0.04

TTE screening may offer significant benefit in this over 50 population, especially those with hypertension, by identifying clinically-relevant structural heart abnormalities, facilitating timely referral, and mitigating the consequences of undiagnosed CVD.

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HUDDLE was a multicity, cross-sectional study that assessed the prevalence of CVD and related risk factors in former NFL players aged ≥50 years through participant questionnaires and noninvasive cardiac screening (BP, ECG, TTE). This study affirmed the role of echocardiographic screening in those with hypertension, because hypertension was independently predictive of structural pathologies on TTE in former professional athletes. The HUDDLE trial also found a high prevalence of unrecognized CVD among this population, findings that suggest that TTE screening may offer significant benefit in individuals aged ≥50 years by identifying clinically relevant structural heart abnormalities, facilitating timely referral, and mitigating the consequences of undiagnosed CVD. *N = 15 participants from the pilot study were excluded; the specific follow-up reason was not collected. †Participants could have multiple abnormal findings warranting follow-up. BP = blood pressure; CVD = cardiovascular disease; ECG = electrocardiogram; HUDDLE = Heart Health: Understanding and Diagnosing Disease by Leveraging Echocardiograms; NFL = National Football League; TTE = transthoracic echocardiography.

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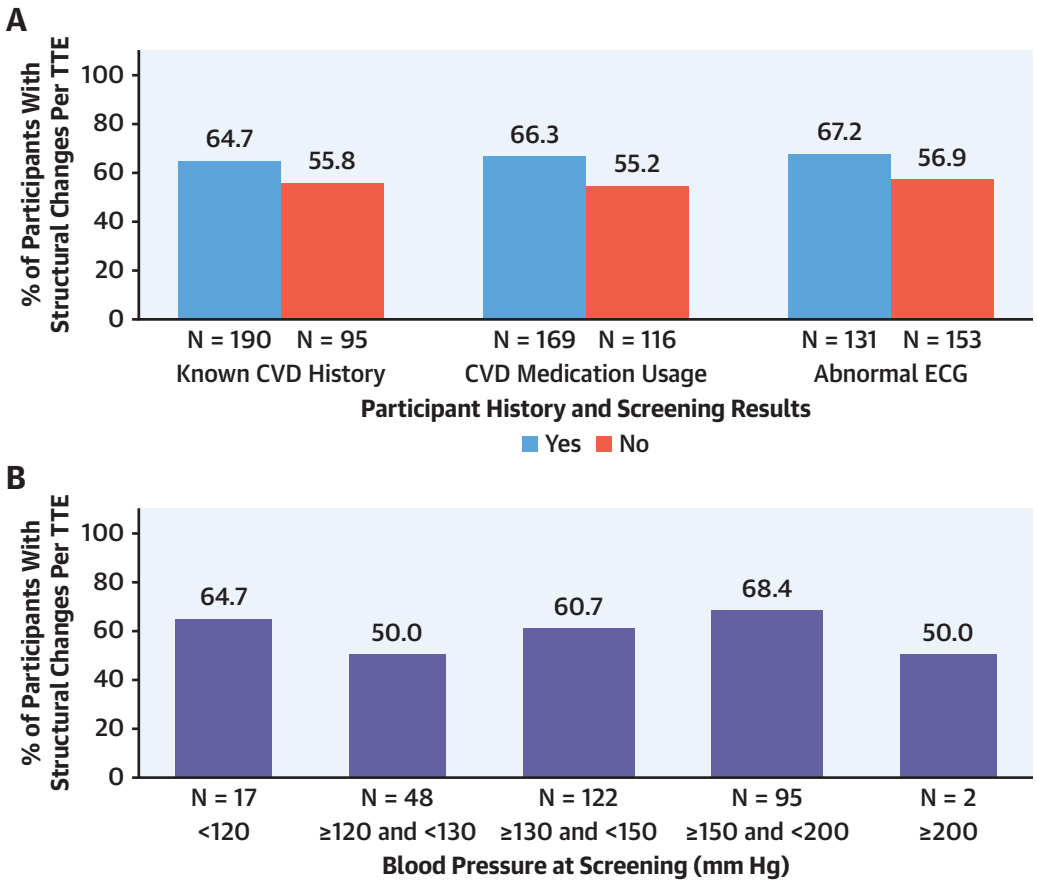
FIGURE 2 Structural Abnormalities Per TTE by Participant History and Screening Results



Structural heart abnormalities per TTE observed in (A) those with or without known CVD history, reported CVD medication use, or abnormal electrocardiogram findings at screening and (B) by BP measurements at screening. Of 11 participants who were normotensive at screening but had structural abnormalities, 8 reported CVD history and/or taking an antihypertensive medication, 2 were on lipid-lowering agents, and 1 reported no CVD history or medications. BP = blood pressure; CVD = cardiovascular disease; ECG = electrocardiogram; TTE = transthoracic echocardiography.

higher among those who reported a known CVD history, reported taking CVD medications, and had an abnormal ECG finding at screening; however, similar rates of structural abnormalities were identified among those who reported no known history of CVD or CVD medication use or who did not have any prognostic ECG findings (**Figure 2A**). A list of observed structural heart abnormalities per TTE is detailed in Supplemental Table 8.

There was a strong association of elevated BP and structural abnormalities in the study population (**Figure 2B**). Elevated BP was prevalent in 93.6% of individuals with an elevated left ventricular mass index (LVMI) (**Figure 3A**). Of the 3 participants who were normotensive with elevated LVMI, 2 reported taking antihypertensive medications, of whom 1 had a history of HF. The third participant did not report any CVD history, although he was reportedly taking rosuvastatin and low-dose aspirin. Similar associations of increasing systolic BP among select structural heart abnormalities suggest pathologic changes (**Figures 3B and 3C**).

INVESTIGATOR-RECOMMENDED FOLLOW-UP. All screened participants were followed up at 30 days (Supplemental Figure 2). Based on a combination of questionnaire results and screening findings, study investigators recommended that 76.7% (n = 207) of

participants seek additional follow-up. The most common reasons for further follow-up recommendation were for elevated BP (56.5%) and abnormal TTE (56.0%) (Central Illustration). Of participants who reported no known history of CVD, 65.6% were recommended for further follow-up because of possible CVD (Figure 4).

PREDICTORS OF STRUCTURAL HEART ABNORMALITIES ON TTE. On multivariable logistic regression modeling, being hypertensive at screening was independently associated with structural heart abnormalities on TTE (OR: 2.02; 95% CI: 1.03-3.96; $P = 0.04$) (Table 3). Age, abnormal ECG findings, and CVD medication use were significant predictors of structural abnormalities at the 0.1 level when tested using univariate analysis; however, they were no longer significant when added to the multivariable model, which may suggest some collinearity.

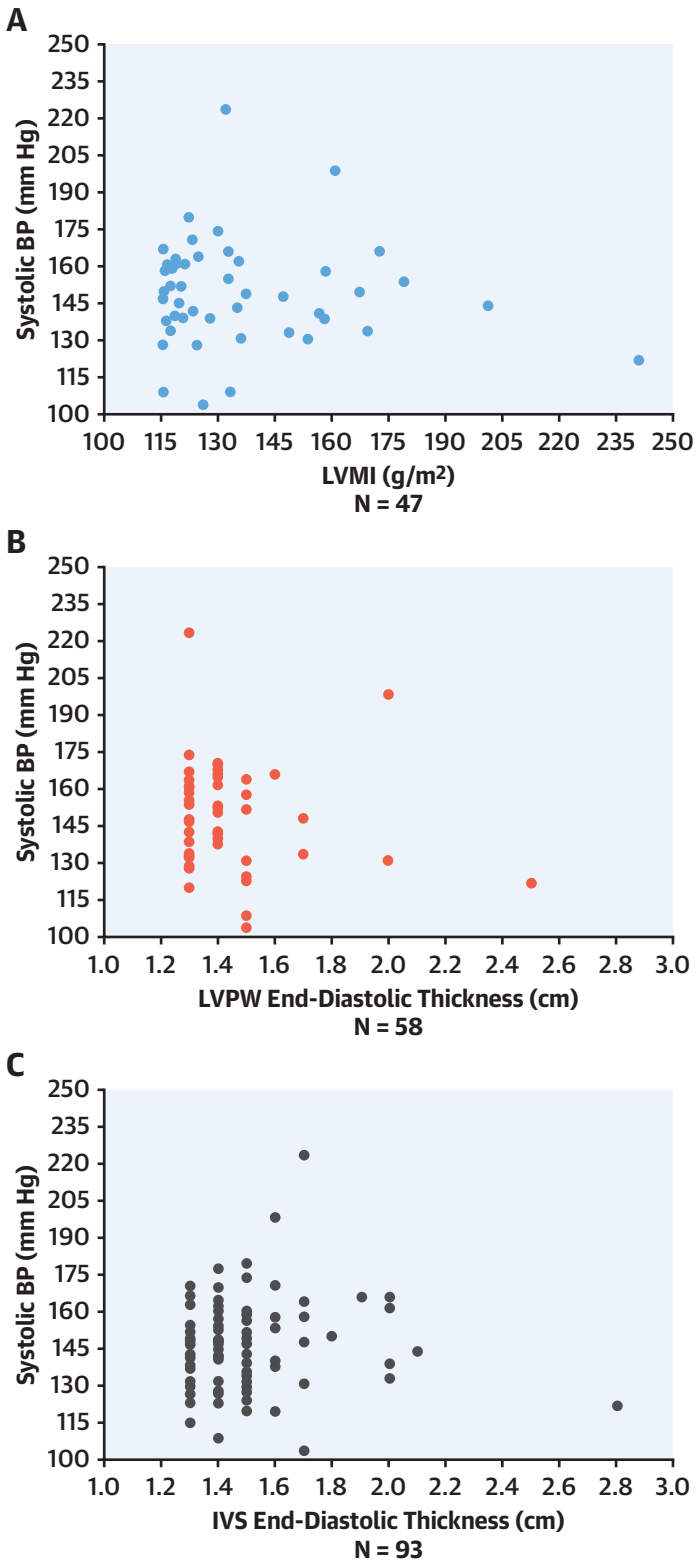
DISCUSSION

The current investigation describes the prevalence of CV risk factors and presence of undiagnosed, clinically relevant structural abnormalities on TTE in former NFL players. The study population comprised 67.6% Black and 26.4% White participants and approximates the racial composition of the NFL, which consisted of 74.3% Black and 24.4% White players between 2014 and 2020.[11]

Our principal findings are 3-fold. First, the prevalence of HTN was established in 89.8% of the population, despite only 37.5% of participants reporting a history of HTN, illustrating a large discrepancy in self-awareness and actual prevalence of CVD and associated risk factors. Second, almost 2 of every 3 screened participants had a structural heart abnormality detected by TTE. Third, HTN at screening was associated with the presence of a clinically relevant structural heart pathology per TTE. To our knowledge, this study represents the first multicity, cross-sectional study of former NFL athletes that examined complementary metrics of CV health and the impact on structural pathologies.

It is well established that, independent of other CV risk factors, uncontrolled HTN is associated with CV morbidity and mortality later in life.[12-14] This is likely explained by the persistent increase in afterload on the left ventricle, which leads to pathologic left ventricle hypertrophy. The association and mechanism of left ventricular hypertrophy and elevated LVMI, a consequence of central pressure overload on the heart from uncontrolled systemic BP, has been well defined.[15-19] In some athletes, a concentric

FIGURE 3 Systolic BP and Structural Findings



Ranges of systolic BP readings at the time of screening are provided for participants who had elevated (A) LVMI, (B) LVPW end-diastolic thickness, and (C) IVS end-diastolic thickness per transthoracic echocardiography. BP = blood pressure; IVS = interventricular septum; LVMI = left-ventricular mass index; LVPW = left-ventricular posterior wall.

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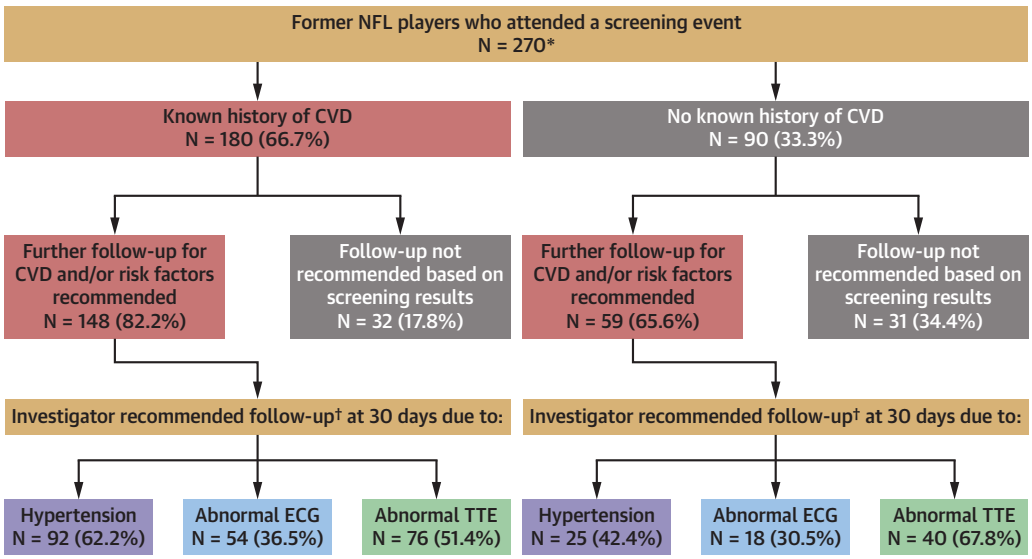
FIGURE 4 Follow-Up Recommendations by Known CVD History



Study investigators followed up with participants 30 days postscreening to review all findings, answer questions, and provide recommendations for additional follow-up based on questionnaires and screening results. *N = 15 participants from the pilot study were excluded because the specific follow-up reason was not collected. †Percentages do not total 100% because participants could have multiple abnormal findings warranting further follow-up. CVD = cardiovascular disease; ECG = electrocardiogram; NFL = National Football League; TTE = transthoracic echocardiogram.

geometric remodeling pattern is typical and has been shown to be independently associated with incident HF, adverse CV outcomes, and increased mortality.[20] We found an unexpectedly high prevalence of HTN among former NFL players. These findings are consistent with previous data that have described the prevalence of HTN to range between 0% to 83%.[21]

With aging, the prevalence of HTN is expected to increase; however, the discrepancy in self-awareness and actual prevalence of HTN highlights a significant knowledge gap in CVD risk factor and management in this population and presents an opportunity for impactful population-based health intervention. In HUDDLE, subjects were scheduled for education sessions on CVD and risk factors as part of the cross-sectional screening. In all sessions, investigators attested to the observed knowledge gap in managing risk factors and identifying signs and symptoms of life-threatening CVD. The interest in knowing more about risk factors and their impact on CVD was confirmed during 30-day follow-up phone calls, where participants expressed enthusiasm in knowing more about and how to follow up for future care. In fact, all study participants successfully completed the follow-up.

The prevalence of subclinical CVD has been established to range between 35% and 55% by several reports and longitudinal studies investigating CV risk factors in communities that are mainly older adults (aged >70 years) of White race.[22,23] It is striking that we observed a similarly high prevalence of clinically relevant structural heart pathologies in this cohort of comparatively younger adults (aged ≥50 years) who are primarily of Black race. The high prevalence of subclinical CV pathology in former athletes is

TABLE 3 Predictors of Structural Changes per Transthoracic Echocardiography[a]

	OR (95% CI)	*P* Value
Age, y (continuous)	1.03 (1.00-1.07)	0.08
Abnormal ECG findings (yes vs no)	1.68 (1.00-2.82)	0.05
CVD medication use (yes vs no)	1.37 (0.80-2.36)	0.25
HTN at screening (yes vs no)	2.02 (1.03-3.96)	0.04

[a]Covariates were chosen for multivariable model inclusion if they were significant at the 0.1 level based on univariate testing. A full list of covariates tested can be found in Supplemental Table 4.
CVD = cardiovascular disease; ECG = electrocardiogram; HTN = hypertension.

consistent with prior reports.[23-25] As noted previously, the presence of structural abnormalities in this population may stem from the greater propensity to develop adverse left ventricular remodeling upon exposure to these risk factors in athletes. Specifically, there is a higher risk of developing HTN later in life as an athlete, which has been associated with CV morbidity among former professional U.S. football players.[20] HTN may be a potential unifying mechanistic factor in the development of this constellation of early-life subclinical CV pathologies.

Demographics, CVD risk factors, and noninvasive CVD screening parameters were found to be associated with clinically relevant, structural abnormalities on TTE. Subjects with structural heart abnormalities were more likely to report a history of CVD or CV medication use or to be found to have a prognostic ECG finding on screening. Furthermore, the presence of elevated BP was significantly associated with a clinically relevant structural abnormality on TTE. These findings are consistent with previous studies recommending that athletes or former athletes with elevated BP receive an ECG and a TTE to assess the effect of long-standing HTN on potential structural heart pathologies, in addition to aggressive risk factor modification and treatment.[26] Current guidelines recommend TTE as a useful tool for initial CV assessment in asymptomatic adults with HTN.[27,28] These recommendations are based on findings from 2 longitudinal cohort studies that showed an independent association between abnormal LVMI demonstrated on TTE and CV mortality in 2,114 asymptomatic patients with HTN.[29,30] In both studies, screening with a TTE was useful in identifying subjects who were at a higher risk of CV mortality over a median follow-up of 3 years. Our findings of an unexpectedly high prevalence of subclinical HTN in this population, who are presumed to be healthy compared to their nonathletic counterparts, call for targeted population health interventions. Specifically, screening high-risk populations such as those with HTN and other cardiometabolic risk factors with a TTE can be effective in identifying structural heart abnormalities and preventing CVD.[3] Large-scale studies are needed to further define populations who would most benefit from early CVD screening and education.

STUDY LIMITATIONS. First, inherent issues exist when using participant-reported data, including recall and social desirability bias. Future studies may benefit from medical record review, especially for medication use. Second, geographic representation was largely from the southern United States, which may affect the generalizability of these findings. Additionally, there was no comparison group, which may also affect generalizability. Variations in recommended follow-up based on individual clinician judgment were also possible. Finally, there was a lack of long-term follow-up because of the cross-sectional nature of this study. Future trials with longer-term follow-up will provide more insight into the benefits of proactive screening for CVD.

CONCLUSIONS

HUDDLE demonstrated a surprisingly high prevalence of HTN among former NFL players and revealed a significant discrepancy between participants' awareness and observed disease prevalence. Structural pathologies observed in this population of former professional athletes were independently predicted by HTN at screening, affirming the role of TTE in those with HTN. Early TTE screening may offer a significant benefit for this population aged older than 50 years by assessing for structural pathologies, facilitating timely referrals, and mitigating the consequences of undiagnosed CVD.

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PERSPECTIVES

COMPETENCY IN SYSTEM-BASED PRACTICE: Among former professional players in the National Football League, the prevalence of unrecognized cardiovascular disease is high. Hypertension predicts echocardiographic pathology in those over 50 years old.

TRANSLATIONAL OUTLOOK: Further efforts are needed to increase awareness of cardiovascular risk among professional football players, to develop strategies to ameliorate risk, to sustain these efforts in the years after retirement from the sport, and to engage in risk reduction initiatives.

REFERENCES

1. McClellan M, Brown N, Califf RM, Warner JJ. Call to action: urgent challenges in cardiovascular disease: a presidential advisory from the American Heart Association. *Circulation*. 2019;139(9):e44–e54. https://doi.org/10.1161/CIR.0000000000000652

2. Vaduganathan M, Mensah GA, Turco JV, Fuster V, Roth GA. The global burden of cardiovascular diseases and risk: a compass for future health. *J Am Coll Cardiol*. 2022;80(25):2361–2371. https://doi.org/10.1016/j.jacc.2022.11.005

3. Tsao CW, Aday AW, Almarzooq ZI, et al. Heart disease and stroke statistics—2023 update: a report from the American Heart Association. *Circulation*. 2023;147(8):e93–e621. https://doi.org/10.1161/CIR.0000000000001123

4. Arnett DK, Blumenthal RS, Albert MA, et al. 2019 ACC/AHA guideline on the primary prevention of cardiovascular disease: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2019;74(10):e177–e232.

5. Naghavi M, Maron DJ, Kloner RA, et al. Coronary artery calcium testing: a call for universal coverage. *Prev Med Rep*. 2019;15:100879. https://doi.org/10.1016/j.pmedr.2019.100879

6. McHugh C, Hind K, Cunningham J, Davey D, Wilson F. A career in sport does not eliminate risk of cardiovascular disease: a systematic review and meta-analysis of the cardiovascular health of field-based athletes. *J Sci Med Sport*. 2020;23(9):792–799. https://doi.org/10.1016/j.jsams.2020.02.009

7. Mitchell C, Rahko PS, Blauwet LA, et al. Guidelines for performing a comprehensive transthoracic echocardiographic examination in adults: recommendations from the American Society of Echocardiography. *J Am Soc Echocardiogr*. 2019;32(1):1–64. https://doi.org/10.1016/j.echo.2018.06.004

8. Kligfield P, Gettes LS, Bailey JJ, et al. Recommendations for the standardization and interpretation of the electrocardiogram: part I: the electrocardiogram and its technology: a scientific statement from the American Heart Association Electrocardiography and Arrhythmias Committee, Council on Clinical Cardiology; the American College of Cardiology Foundation; and the Heart Rhythm Society: endorsed by the International Society for Computerized Electrocardiology. *Circulation*. 2007;115(10):1306–1324. https://doi.org/10.1161/CIRCULATIONAHA.106.180200

9. Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2018;71(19):2199–2269. https://doi.org/10.1016/j.jacc.2017.11.005

10. Lang RM, Badano LP, Mor-Avi V, et al. Recommendations for cardiac chamber quantification by echocardiography in adults: an update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. *J Am Soc Echocardiogr*. 2015;28(1):1–39.e14. https://doi.org/10.1016/j.echo.2014.10.003

11. Marquez-Velarde G, Grashow R, Glass C, et al. The paradox of integration: racial composition of NFL positions from 1960 to 2020. *Sociol Race Ethnic*. 2023;9(4):451–469. https://doi.org/10.1177/23326492231182597

12. McCarron P, Smith GD, Okasha M, McEwen J. Blood pressure in young adulthood and mortality from cardiovascular disease. *Lancet*. 2000;355(9213):1430–1431. https://doi.org/10.1016/S0140-6736(00)02146-2

13. Pletcher MJ, Bibbins-Domingo K, Liu K, et al. Nonoptimal lipids commonly present in young adults and coronary calcium later in life: the CARDIA (Coronary Artery Risk Development in Young Adults) study. *Ann Intern Med*. 2010;153(3):137–1346. https://doi.org/10.7326/0003-4819-153-3-201008030-00004

14. Baker JL, Olsen LW, Sorensen TI. Childhood body-mass index and the risk of coronary heart disease in adulthood. *N Engl J Med*. 2007;357(23):2329–2337. https://doi.org/10.1056/NEJMoa072515

15. Cuspidi C, Sala C, Negri F, Mancia G, Morganti A, Italian Society of Hypertension. Prevalence of left-ventricular hypertrophy in hypertension: an updated review of echocardiographic studies. *J Hum Hypertens*. 2012;26(6):343–349. https://doi.org/10.1038/jhh.2011.104

16. Ghosh AK, Francis DP, Chaturvedi N, et al. Cardiovascular risk factors from early life predict future adult cardiac structural and functional abnormalities: a systematic review of the published literature. *J Cardiol Ther*. 2014;2(2):78–87. https://doi.org/10.12970/2311-052X.2014.02.02.4

17. Lai CC, Sun D, Cen R, et al. Impact of long-term burden of excessive adiposity and elevated blood pressure from childhood on adulthood left ventricular remodeling patterns: the Bogalusa Heart Study. *J Am Coll Cardiol*. 2014;64(15):1580–1587. https://doi.org/10.1016/j.jacc.2014.05.072

18. Yan Y, Li S, Guo Y, et al. Life-course cumulative burden of body mass index and blood pressure on progression of left ventricular mass and geometry in midlife: the Bogalusa Heart Study. *Circ Res*. 2020;126(5):633–643. https://doi.org/10.1161/CIRCRESAHA.119.316045

19. Savage DD, Garrison RJ, Kannel WB, et al. The spectrum of left ventricular hypertrophy in a general population sample: the Framingham Study. *Circulation*. 1987;75(1 pt 2):I26–I33.

20. Drazner MH. The progression of hypertensive heart disease. *Circulation*. 2011;123(3):327–334. https://doi.org/10.1161/CIRCULATIONAHA.108.845792

21. Berge HM, Isern CB, Berge E. Blood pressure and hypertension in athletes: a systematic review. *Br J Sports Med*. 2015;49(11):716–723. https://doi.org/10.1136/bjsports-2014-093976

22. Fang M, Wang D, Tang O, et al. Subclinical cardiovascular disease in US adults with and without diabetes. *J Am Heart Assoc*. 2023;12(11):e029083. https://doi.org/10.1161/JAHA.122.029083

23. Shah AM, Claggett B, Loehr LR, et al. Heart failure stages among older adults in the community: the Atherosclerosis Risk in Communities Study. *Circulation*. 2017;135(3):224–240. https://doi.org/10.1161/CIRCULATIONAHA.116.023361

24. Kim JH, Hollowed C, Liu C, et al. Weight gain, hypertension, and the emergence of a maladaptive cardiovascular phenotype among US football

players. *JAMA Cardiol*. 2019;4(12):1221–1229. https://doi.org/10.1001/jamacardio.2019.3909

25. Kim JH, Hollowed C, Patel K, et al. Temporal changes in cardiovascular remodeling associated with football participation. *Med Sci Sports Exerc*. 2018;50(9):1892–1898. https://doi.org/10.1249/MSS.0000000000001631

26. Tso JV, Kim JH. Hypertension in athletes: clinical implications and management strategies. *Cardiol Clin*. 2023;41(1):15–24. https://doi.org/10.1016/j.ccl.2022.08.002

27. American College of Cardiology Foundation Appropriate Use Criteria Task Force, American Society of Echocardiography, American Heart Association, et al. ACCF/ASE/AHA/ASNC/HFSA/HRS/SCAI/SCCM/SCCT/SCMR 2011 appropriate use criteria for echocardiography. a report of the American College of Cardiology Foundation Appropriate Use Criteria Task Force, American Society of Echocardiography, American Heart Association, American Society of Nuclear Cardiology, Heart Failure Society of America, Heart Rhythm Society, Society for Cardiovascular Angiography and Interventions, Society of Critical Care Medicine, Society of Cardiovascular Computed Tomography, and Society for Cardiovascular Magnetic Resonance endorsed by the American College of Chest Physicians. *J Am Coll Cardiol*. 2011;57(9):1126–1166. https://doi.org/10.1016/j.jacc.2010.11.002

28. Greenland P, Alpert JS, Beller GA, et al. 2010 ACCF/AHA guideline for assessment of cardiovascular risk in asymptomatic adults: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *J Am Coll Cardiol*. 2010;56(25):e50–e103.

29. Verdecchia P, Carini G, Circo A, et al. Left ventricular mass and cardiovascular morbidity in essential hypertension: the MAVI study. *J Am Coll Cardiol*. 2001;38(7):1829–1835. https://doi.org/10.1016/s0735-1097(01)01663-1

30. Rodriguez CJ, Lin F, Sacco RL, et al. Prognostic implications of left ventricular mass among Hispanics: the Northern Manhattan Study. *Hypertension*. 2006;48(1):87–92. https://doi.org/10.1161/01.HYP.0000223330.03088.58

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APPENDIX For supplemental tables and figures, please see the online version of this paper.